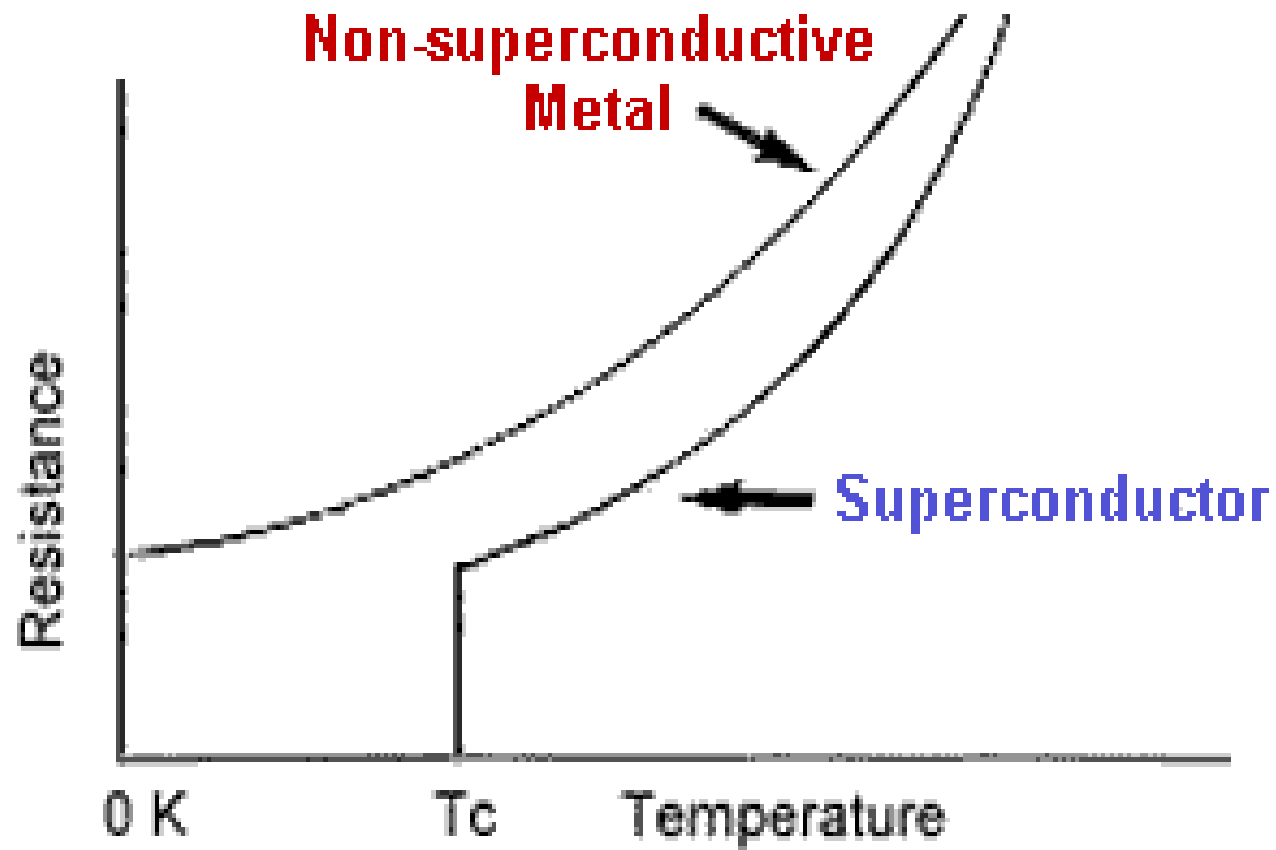


Superconductors

Syllabus: Temperature dependence of resistivity in superconducting materials, effect of magnetic field (Meissner effect), Temperature dependence of critical field, London equations, Josephson theory, persistent current, Type I and Type II superconductors, BCS theory (Qualitative), High temperature superconductors and Applications of superconductors.

Superconductivity

- The phenomena of disappearance of electrical resistance of some materials or compounds below a certain temperature were called superconductivity. It was first observed by K. Onnes in 1911 in Mercury at 4K.
- The materials or compounds exhibiting this property are known as superconductors.
- The temperature, at which the transition from normal state to super conducting state takes place on cooling the material, in the absence of magnetic field, is called the critical temperature (T_c) or the transition temperature.
- The superconducting state is a distinct phase of matter having characteristic electrical, magnetic, thermodynamic and other physical properties.



Persistent Current

- If a superconductor has the shape of a ring, a current can be induced in it by electromagnetic induction and simply cool the ring in a magnetic field from a normal temperature T to below T_c and then remove the field.
- Now, it has been observed that this current continues to persist with undiminished strength for days.
- Such currents are called persistent current and are quite expected in superconductors.
- The complete disappearance of resistivity may be demonstrated by making use of the property of conducting ring and electromagnetic induction.

Consider a ring exposed to an external magnetic field or flux density B which is changing as a function of time. The current $I(t)$ flowing in the ring at time t is given by Lenz's law which states that

$$-A \frac{dB}{dt} = RI(t) + L \left[\frac{dI(t)}{dt} \right]$$

Where A is area of ring, R is its resistance and L is self inductance of the ring. Now if $B = 0$ then,

$$RI(t) + L \left[\frac{dI(t)}{dt} \right] = 0$$

The general solution is

$$I(t) = I(0)e^{-\frac{Rt}{L}}$$

Thus, the current in the ring decays exponentially in the absence of an external changing magnetic field and it eventually vanishes.

However, if the ring becomes superconductor,

then, $R=0$ and $I(t)=I(0)$.

In other words, the current will not decay with time. This current is called persistent current and circulation of the current will give rise to a magnetic field. This property is very useful for high field magnets.

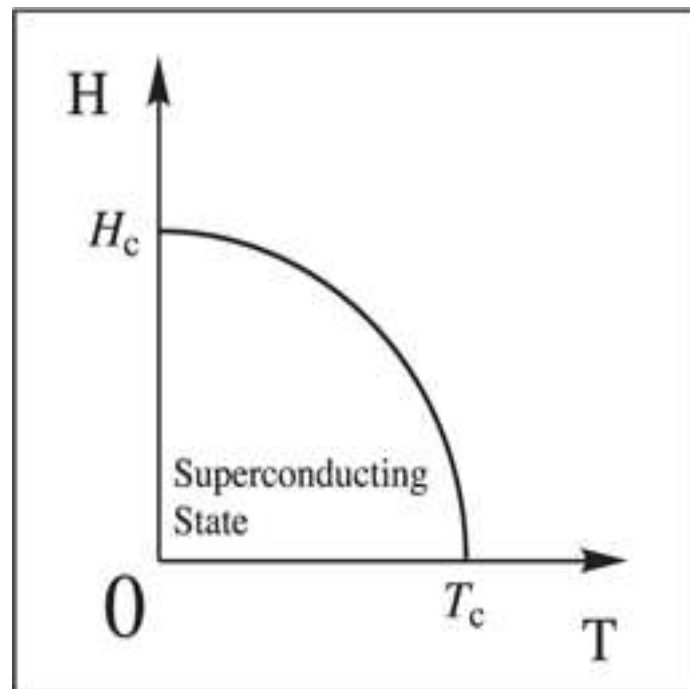
Temperature dependence of critical magnetic field

- Superconducting state of a material mainly depends on temperature and strength of the magnetic field in which it is placed .
- Superconductivity disappears if the temperature of the specimen is raised above T_c
- Superconductivity also disappears if the a strong enough magnetic field is applied.
- At temperature below T_c , in the absence of any magnetic field, the material is in superconducting state.

- When the strength of magnetic field applied reaches a critical value H_c , the superconductivity disappears.
- At $T=T_c$, $H_c=0$. At temperatures below T_c , H_c increases. The dependence of the critical field upon the temperature is given by

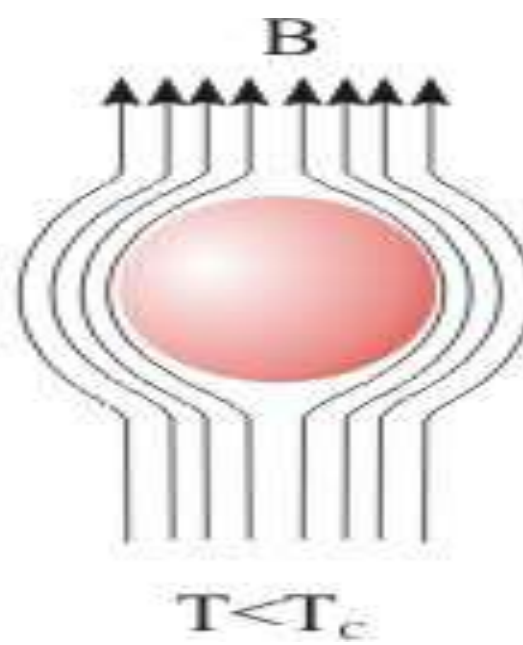
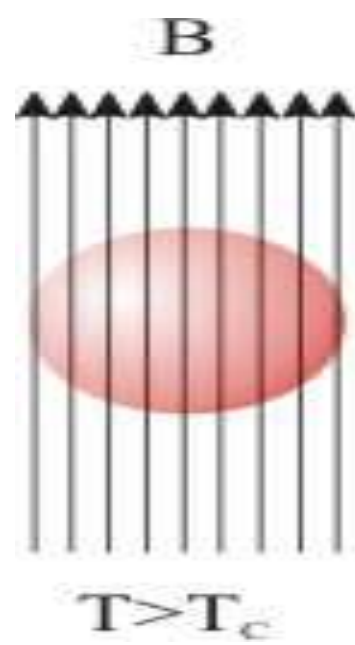
$$H_c = H_c(0) \left[1 - \left\{ \frac{T}{T_c} \right\}^2 \right]$$

- Where $H_c(0)$ is the critical field at 0K. H_c and T_c are constants and characteristics of the material.
- Variation of H_c with temperature given in the figure.



Meissner effect

- Meissner and Ochselfeld in 1933 found that if a superconductor is cooled in magnetic field below to its critical temperature corresponding to that field, then the lines of induction B are pushed out of the body of the superconductor at the transition this phenomenon is called the “Meissner Effect”.
- Such flux exclusion is also observed if the superconductor is first cooled below T_c and then placed in the magnetic field, i.e. the effect is vice versa or, reversible. The effect is of great importance.



As magnetic induction B produced in any material is

$$B = \mu_0 (H + M) = \mu_0 (1 + \chi) H$$

Since $B=0$ in superconducting state

$M = -H$, $\chi = -1$ and also $\chi = \mu_r - 1$ means $\mu_r = 0$

i.e, a superconductor exhibits perfect diamagnetism. Because of diamagnetic nature, superconducting materials strongly repel external magnets. It leads to a levitation effect.

Perfect diamagnetism of superconductor and zero resistivity are independent property

- From Ohm's law $E = \rho j$
- We see that if the resistivity goes to zero while j is held finite, the E must be zero.
- Using Maxwell's third equation, we have $B = \text{const.}$
- This concludes that the flux through the perfect conductor cannot change on cooling through the transition.
- This means that when a perfect conductor is cooled in the magnetic field until its resistance becomes zero, the magnetic field in the material gets frozen in and cannot change subsequently irrespective of the applied field.
- This is in contradiction with the Meissner effect.
- Thus, perfect diamagnetism and zero resistivity are the two independent essential properties of the superconducting state.

Effect of Current on Superconducting state (Critical Current I_c)

- The critical magnetic field required to destroy superconductivity need not necessarily be applied externally.
- An electric current flowing through the superconducting material itself may give rise to necessary magnetic field.
- For example, when a current flows through a superconducting ring, it gives rise to its own magnetic field.
- As the current is increased to a critical value I_c , the associated magnetic field becomes H_c and the superconductivity disappears.
- The critical current I_c flowing through a superconducting ring of radius r is given by

$$I_c = 2\pi r H_c$$

- This limits the maximum possible current that flows through superconductor.
- Hence, this is the main hurdle in producing high field superconducting magnets.

Critical Current (Silsbee's Rule)

The minimum value of current that can be passed in a sample of superconductor without destroying the superconductivity is defined as critical current.

If a current I is flowing through a wire of radius r of a soft superconductor, then the surface magnetic field H_s associated with current is

$$H_s = I/(2\pi r)$$

In addition to this internal field (surface field), if an external magnetic field is also present, then to sustain the superconducting state the total magnetic field strength should be less than or equal to the critical field, i.e.,

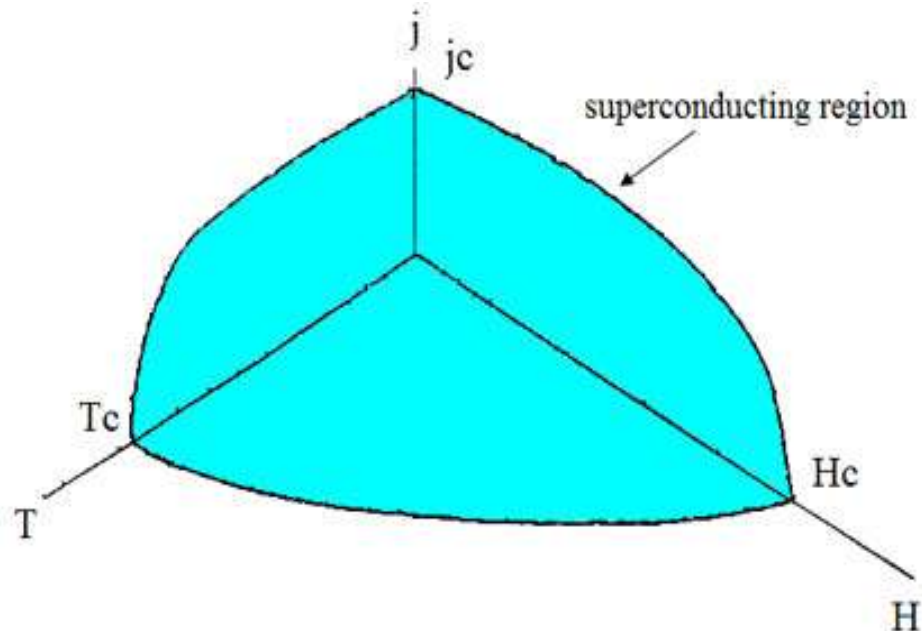
$$H_c = H_s + 2H$$

Or, $H_s = H_c - 2H$

Or in terms of critical current

$$I_c = 2\pi r (H_c - 2H)$$

This is called Silsbee's Rule.



Isotope Effect

- In superconducting materials, the transition temperature varies with average isotopic mass M , of their constituents.
- The variation is found to follow the general form

$$T_c \propto M^{-\alpha}$$

or,

$$T_c M^{\alpha} = \text{Constant}$$

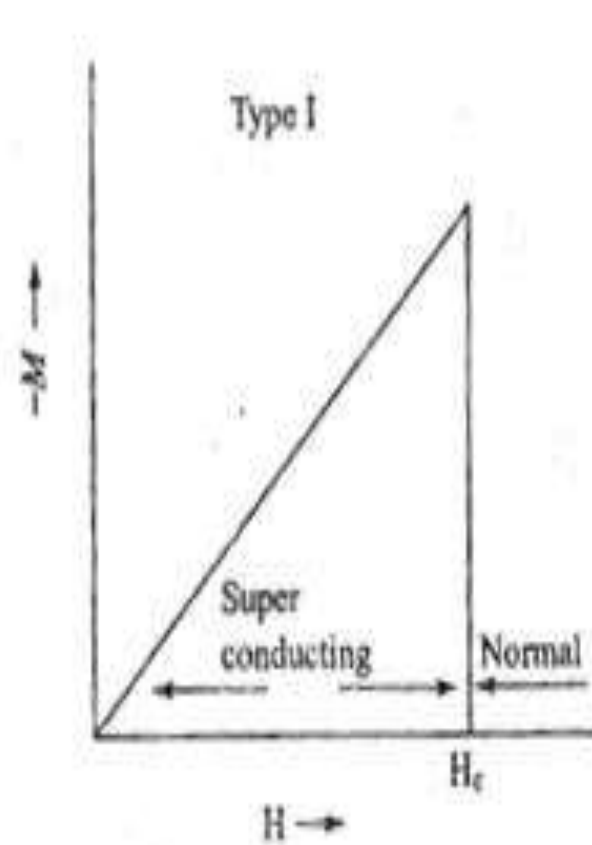
where α is called the isotope effect coefficient and is defined as

$$\alpha = \frac{\partial \ln T_c}{\partial \ln M}$$

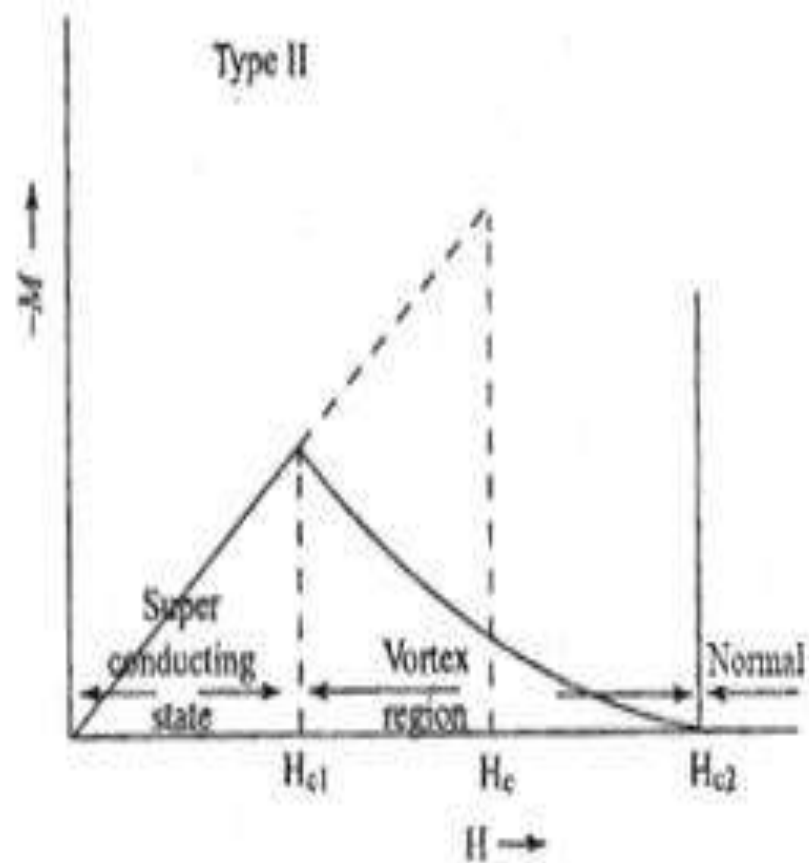
In most cases $\alpha = 0.5$.

Type I and Type II Superconductors

- Based on diamagnetic response superconductors can be classified as type I and type II superconductors.
- Superconductors exhibiting a complete Meissner effect (perfectly diamagnetism) are called type I superconductors also known as soft superconductors.
- When the magnetic field strength is gradually increased from its initial value $H < H_c$, at H_c the diamagnetism abruptly disappears and the transition from superconducting state to normal state is sharp as shown in figure.



(a)



(b)

- Pure specimen of Al, Zn, Hg and Sn are some examples of type I superconductors.
- Type II superconductors, as shown in second figure, upto field H_{c1} the specimen is in a pure superconducting state. The magnetic flux lines are rejected. When the field is increased beyond H_{c1} (the linear critical field, the magnetic flux lines starts penetrating. The specimen is in a mixed state between H_{c1} and H_{c2} (upper critical field). Above H_{c2} , the specimen is in a normal state.
- This means that the Meissner effect is incomplete in the region between H_{c1} and H_{c2} . This region is known as Vortex-region.
- Type II superconductors are known as hard superconductors. Zr, Nb and ceramics are examples of this type.
- Type II superconductors are of great practical interest because of the high current densities that they can carry.

BCS- theory

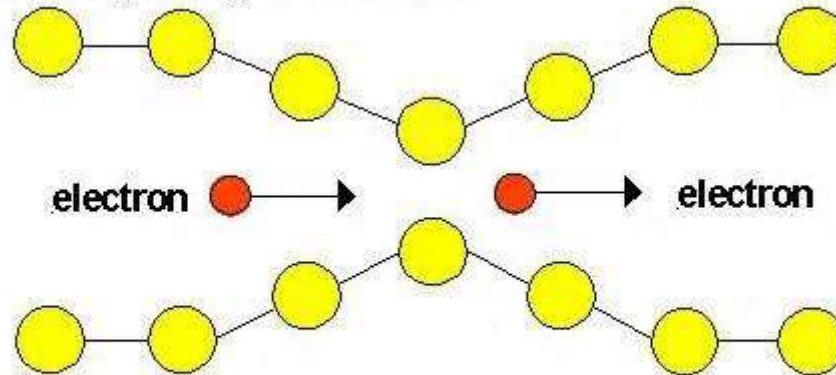
(Explanation for the occurrence of superconductivity)

BCS Theory of superconductivity was put forward by Bardeen, Cooper and Schrieffer in 1957 and hence named as BCS theory. This theory could explain superconducting nature. It is based on advanced quantum concept.

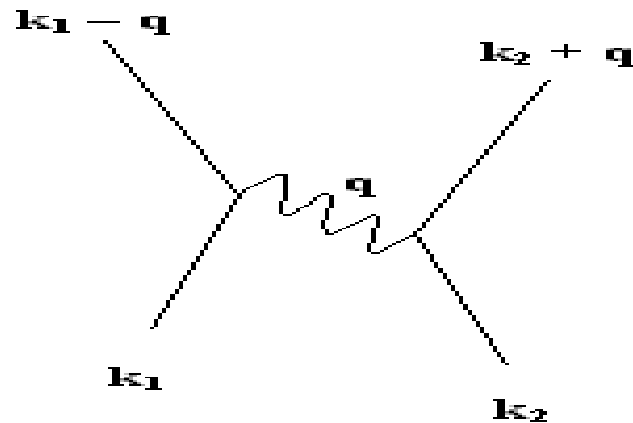
(i) Electron-electron interaction via lattice deformation:

- Consider an electron passing through the lattice of positive ions. The electron is attracted by the neighboring positive ions. It forms a positive ion core as shown in figure and gets screened by them.
- The screening greatly reduces the effective charge of this electron. Due to this attraction between the electron and ion core, the lattice get deformed on local scale.
- Now if another electron passes by the side of the assembly of the said electron and the ion core, it gets attracted towards the assembly.

positively charged lattice ions



Cooper pair moving through lattice



- The second electron interacts with the first electron via lattice deformation.
- This interaction is said to be due to the exchange of a virtual phonon ‘ q ’, between the two electrons.

- The interaction process can be written in terms of the wave vectors k as

$$k_1 - q = k_1' \text{ and } k_2 + q = k_2'$$

- This gives $k_1 + k_2 = k_1' + k_2'$, i.e., the net wave vector of the pair is conserved.
- The momentum is transferred between the electrons.
- These two electrons together form a Cooper pair and is known as Cooper electron.

(ii) Cooper Pair:

- Cooper showed that if there is an attraction between two electrons, they are able to form a bound state so that their total energy is less than $2E_F$.
- These electrons are paired to form a single system. Their motions are correlated.
- The binding is strongest when the electrons forming the pair have opposite momenta and opposite spins, i.e., $k\uparrow$, and $-k\downarrow$.

All electron pairs with attraction among them and lying in the neighbourhood of the Fermi surface form cooper pairs.

- In superconducting material, when a pair of electrons lying just below the Fermi surface is taken just above it, they form a cooper pair and their total energy is reduced.
- This continues until the system can gain no additional energy by pair formation.
- Thus the total energy of the system is further reduced.
- At absolute zero in superconductors there is no abrupt discontinuity in the distribution of electrons across the Fermi surface.
- These cooper electrons or superelectrons are responsible for the zero resistivity at low temperatures.

Josephson Effect

- Consider a thin insulating layer ($20 \text{ \AA}$) sandwiched between two metals.
- This insulating layer acts as a potential barrier for flow of electron from one metal to another.
- Now consider thin insulating layer sandwiched between two superconductors.
- The superelectrons (cooper pairs) tunnel through the
- insulating layer from one superconductor to another without dissociation, even at zero potential difference across the junction.
- Their wavefunctions on both sides are highly correlated. This is known as Josephson effect.

London Equations

- London equations are quantitative explanation of Meissner effect and zero resistivity, given by brothers F. London and H. London in 1935, by incorporating above two property to Maxwell's equations.
- Consider a superconductor composed of superfluid and normal electrons. Let their densities are n_s and n_n and their velocities are V_s and V_n .
- So, total no. of electrons are given by

$$n_0 = n_s + n_n$$

- The equation of motion for superfluid electron is

$$m \frac{dV_s}{dt} = -eE$$

$$J_s = -en_s V_s$$

$$\frac{dJ_s}{dt} = \frac{n_s e^2}{m} E$$

This is known as First London Equation.

Taking Curl of above equation

$$\nabla \times \frac{dJ_s}{dt} = \frac{n_s e^2}{m} \nabla \times E$$

Using third Maxwell Equation

$$\nabla \times \frac{dJ_s}{dt} = -\frac{n_s e^2}{m} \cdot \frac{\partial B}{\partial t}$$

Which after integration with respect to time yields

$$\nabla X J_s = - \frac{n_s e^2}{m} \cdot B$$

This is known as second London equation and is only correct for superconductors.

From fourth Maxwell equation

$$\nabla X B = \mu_0 J_s$$

or,

$$\nabla X (\nabla X B) = \mu_0 \nabla X J_s$$

or,

$$-\nabla^2 B = -\mu_0 \frac{n_s e^2}{m} B$$

or,

$$\nabla^2 B = \frac{1}{\lambda^2} B$$

Where λ is called the London penetration depth and is defined as

$$\lambda = \left(\frac{m}{\mu_0 n_s e^2} \right)^{\frac{1}{2}}$$

The magnetic field does not stop abruptly at the superconductor surface but penetrates with exponential attenuation into the material.

High Temperature Superconductors

- Any superconductor with a transition temperature above 30K is in general called high temperature superconductors. The high T_c superconductors developed so far belong to five chemical systems having the following general formula-
- 1- $\text{Ba Pb}_{1-x} \text{Bi}_x \text{O}_3$
- 2- $\text{La}_{2-x} \text{M Cu O}_{4-x}$ (M is Ba or Sr)
- 3- $\text{Ba}_2 \text{M}_x \text{Cu}_3 \text{O}_{7-x}$ (M is Y or rare earth metals such as Gd ,Eu etc.)
- 4- $\text{Ba}_{2-x} \text{La}_{1+x} \text{Cu}_3 \text{O}_\delta$
- 5- $\text{Bi}_2 \text{Ca Sr}_2 \text{Cu}_2 \text{O}_\delta$

- It is difficult to state the mechanism of high T_c superconductivity. It has been realized that role oxygen is essential for high T_c oxide superconductors.
- The unit cell of high T_c superconductors has complicated structure. If the cell contains one atom of rare earth metal, two Barium atoms, three copper atoms and seven oxygen atoms, then such compounds are named 1-2-3 superconductors.
- Certain high T_c superconductors do not have the same crystal structure as 1-2-3. Most of the compounds are layered copper oxide materials.
- Some characteristics of high T_c superconductors are as follows:
 - 1- Highly anisotropic and have layered structures.
 - 2- Metallic oxides.
 - 3- Ceramic materials.

Applications of Superconductors

- These materials are used for producing very strong magnetic fields of about 50 Tesla, which is much larger than the field obtainable from an electromagnet.
- High current densities with zero resistance properties of superconducting materials make useful for strong electromagnets for example, in MRI devices used in medicine.
- In superconducting materials, heating loss is zero, therefore power can be transmitted through superconducting cables without loss.
- These materials can be used to perform fast logic and storage functions in computers.

- Type II superconductors are mainly utilized for superconducting solenoids.
- These are also used in high speed levitated trains (Maglev).
- SQUIDs (Superconducting Quantum Interface Devices) are used in field of medicine, it measures the very weak fields generated by heart (10^{-10} T) and brain (10^{-13} T).
- Recently superconductors have found applications in switching elements called cryotrons.
- These materials are used in RF and microwave filters (e.g., for mobile phone base stations).
- BSCCO-2212 is the first high T_c superconductor to be used for making conducting wires.
- In 1997 Swiss-Swedish company (ABB) developed superconductor based transformers and fault limiters.
- Superconducting Y-Ba-Cu-O is used in antenna applications.